

Practice Quiz 05B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 05

Contrapositive or contradiction; a universally quantified goal.

Write in full sentences. Introduce the arbitrary element and name your proof strategy.

1. Prove

$$\forall x \in \mathbf{Z} \ (9 \nmid x^2 \rightarrow 3 \nmid x).$$

Solution Let x be an arbitrary integer. We prove the contrapositive

$$3 \mid x \longrightarrow 9 \mid x^2.$$

Suppose $3 \mid x$. Then $x = 3k$ for some integer k . Therefore

$$x^2 = (3k)^2 = 9k^2.$$

Because k^2 is an integer, $9 \mid x^2$. This proves the contrapositive, so $9 \nmid x^2 \rightarrow 3 \nmid x$. Since x was arbitrary, the universally quantified statement follows. ■